Exp 3

Three - Layer Neural Network for Handwritten Digit Recognition

AIM:

To design and implement a three-layer neural network using backpropagation to recognize handwritten digits from the MNIST dataset.

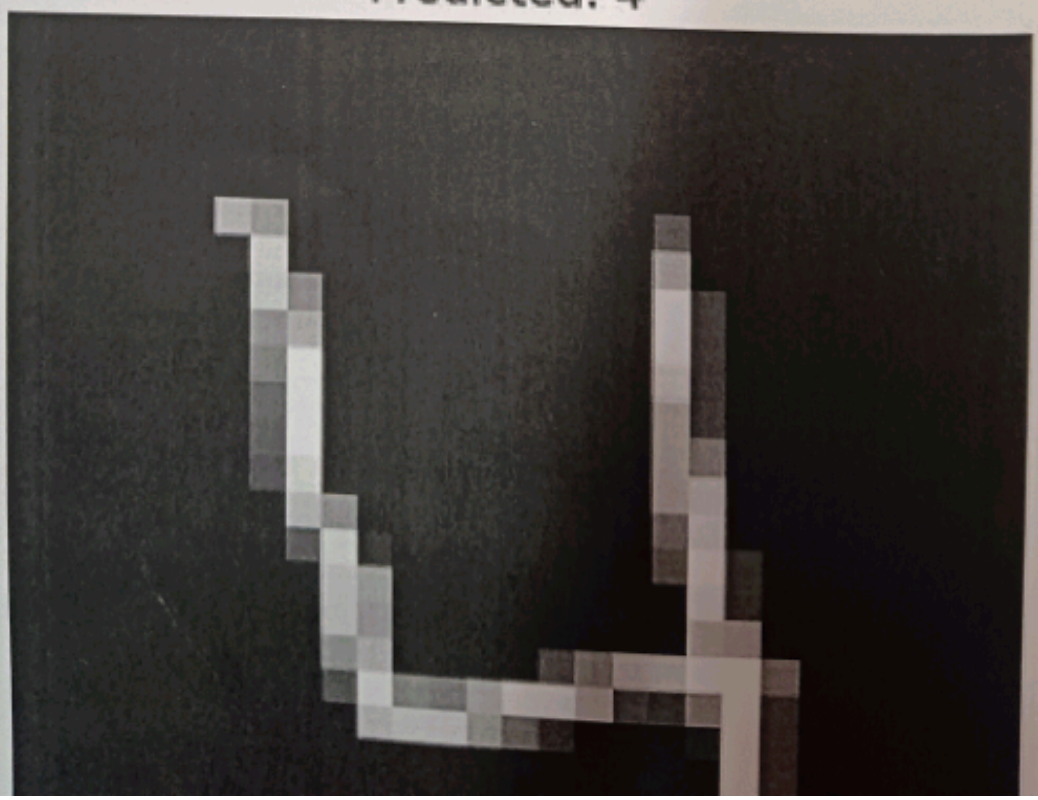
ALGORITHM:

* Load the MNIST training dataset.
* Import required libaries.
* Separate the input images (x) and labels (y).
* Normalize pixel values and reshape images to 28 × 28 × 1.
* Convert class labels into one-hot encoded format.
* Split the dataset into training and validation sets.
* Build a three-layer neural network.
* Compile the model using the Adam optimizer and categorical cross-entropy loss.
* Train the model using the backpropagation algorithm.
* Load and preprocess the test dataset.
* Predict the handwritten digits and display

Predicted: 1



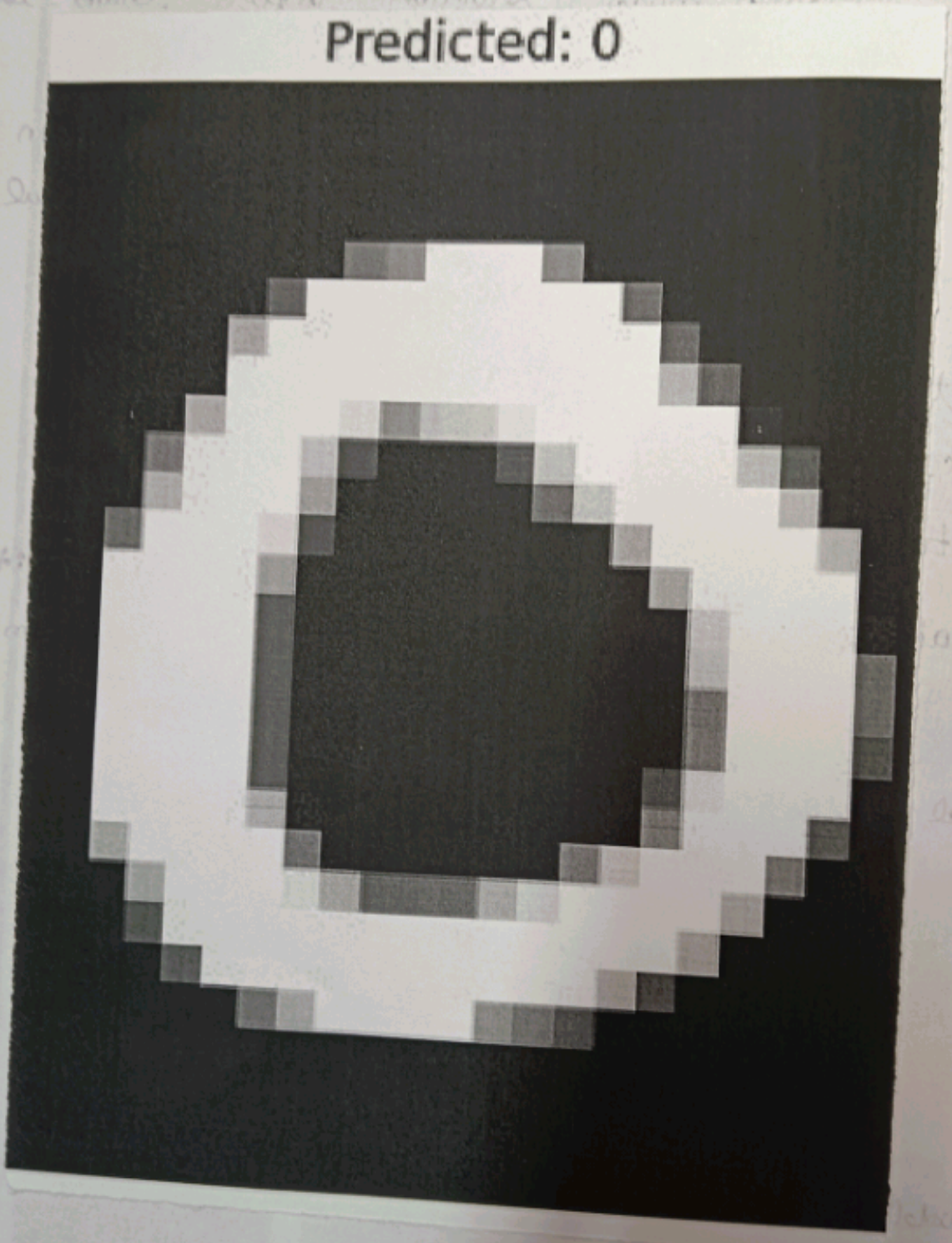
Predicted: 4



CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from tensorflow.keras.models import sequential
from tensorflow.keras.layers import Dense, flatten
from tensorflow.keras.utils import to_categorical
train = pd.read_csv('/content/train.csv')
x = train.iloc[:,1:].values/255.0
x = x.reshape(-1,28,28,1)
y = to_categorical(train.iloc[:,0],10)
x_train, x_val, y_train, y_val = train_test_split(x,y,
                                  test_size=0.2, random_state=42)

model = sequential([
    Input(shape=(28,28,1)),
    Flatten(),
    Dense(128, activation='relu'),
    Dense(64, activation='relu'),
    Dense(10, activation='softmax')
])
model.compile(optimizer='adam', loss='categorical_
                                  crossentropy', metrics=['accuracy
test = pd.read_csv('/content/test.csv')
x_test = test.values/255.0
x_test = x_test.reshape(-1,28,28,1)
```


Predicted: 0

```
pred = model.predict(x_test)
labels = np.argmax(pred, axis=1)
for i in range(5):
    plt.imshow(x_test[i].reshape(28,28), cmap='gray')
    plt.title(f"Predicted: {labels[i]}")
    plt.axis("off")
    plt.show()
```

RESULT:

Thus, the three layer neural network was successfully trained using the MNIST dataset, and it accurately recognized handwritten